Supporting Information for: When Mass Security Screening Backfires: A Signal Detection and Structural Causal Analysis

S1. Resource reallocation: investigative capacity under false positive burden

The classification pathway's prodigious false positive output requires prodigious investigative resources to attempt to sort. This creates resource reallocation effects.

Proposition 1 (Resource tractability of known-CSAM scanning).

Hash-matching of known CSAM achieves $Sp \to 1$ by design. Under these conditions, false positives approach zero and total flags scale with true prevalence: Flags $\approx N \cdot \pi \cdot Se$. The resource burden is therefore bounded by the actual prevalence of known material in the corpus — a finite quantity, manageable through proportional resource allocation.

Corollary 1 (Resource intractability of unknown-CSAM mass scanning).

For any probabilistic classifier with $Sp < 1$ operating at scale N, the minimum specificity required to remain within budget R satisfies $Sp^* = 1 - R/(c \cdot N \cdot (1-\pi))$, which approaches 1 as $N \to \infty$. Since $Sp = 1$ is unachievable for any classifier inferring unknown content from probabilistic signals, no finite investigative budget R satisfies the resource constraint at mass scale. The false positive flood is not an engineering problem solvable by improving classifier accuracy; it is a structural consequence of operating a probabilistic classifier at population scale under rarity. This is what the capacity calculations below instantiate numerically.

Current investigative architecture (two layers). Once a report reaches national law enforcement, there is no lightweight triage stage: police must conduct a full investigation before prosecutorial evaluation. Then, the public prosecutor's office evaluates the case and, where there is sufficient evidence, prepares an indictment. Both layers may already be operating at or near capacity. For example, ZAC NRW, one of Germany's leading cybercrime units, employs 13 prosecutors handling approximately 14,000 cases per year (Hartmann, expert interview, May 2025).

Proposed architecture under Chat Control (three layers). The draft EU Chat Control regulation proposes an additional institutional layer upstream of national law enforcement: a new EU Centre on Child Sexual Abuse. The Commission's description of the EU Centre states it would perform substantive false positive review before forwarding to national authorities — receiving automated reports from service providers, conducting technical verification, and routing confirmed reports to national law enforcement (European Commission, n.d.; Regulation COM (2022) 209 final, Arts. 49-54). This EU Centre would function as a triage and coordination body before cases reach the existing national two-layer pipeline.

Because the three-layer architecture adds an upstream EU Centre as a buffer before national law enforcement, modeling it is the *more generous* test for program proponents. If Chat Control's volume overwhelms even this buffered first layer, then the existing two-layer system — which would absorb the full volume without that buffer — would fare *worse still*. It is therefore logically unnecessary to model the two-layer case separately.

As a rule, this simulation intentionally uses assumptions that are notably generous to Chat Control proponents, following Fienberg's approach of adopting assumptions which were notably favorable to polygraph proponents. The 30-second-per-flag triage assumption applies to this EU Centre layer only, and is an extremely conservative estimate for review of potentially illegal material. Under this generous assumption, EU Centre triage alone consumes millions of person-hours annually (see calculations below).

If cases forwarded from the EU Centre to national police then still require full forensic investigation, as under the current system, then the resource cost per flag is substantially higher than the triage estimate, rendering the overload figures below quite conservative.

Europol IOCTA (2023) estimates approximately 3,000 law enforcement officers across EU member states engaged in CSAM-related work; this serves as the primary capacity estimate. As a corroborating example, ZAC NRW's 13 specialist prosecutors handling ~14,000 cases annually suggests that even a generous scaling to EU population would not materially change the order of magnitude of the specialist bottleneck (Hartmann, expert interview, May 21, 2025). Institutional heterogeneity across EU member states limits the validity of direct population scaling (Hartmann, expert interview, March 6, 2026). Were the resource overload figures under generous assumptions not so overwhelming, this would warrant further research.

The pre-existing strain on this infrastructure is substantial: NCMEC received approximately 36.2 million CyberTipline reports in 2023 (NCMEC 2024), and the Internet Watch Foundation confirmed 275,652 CSAM URLs the same year (IWF 2024) — before Chat Control adds any volume. Even under the generous 3,000-figure capacity and assuming only 30 seconds per flag for EU Centre triage, 1.598 billion false positive flags per 10 billion messages represents approximately 13.3 million person-hours — roughly 2.2× total available capacity.

The resource cost of false positives is not hypothetical. In NRW alone, approximately 2,300 investigations against innocent people were opened and subsequently closed in 2024 following a wave of Facebook account hijackings used to post CSAM (Hartmann, expert interview, May 21, 2025). Each required investigative resources — search warrants, cloud data requests, prosecutorial review — before innocence could be established. This is the current baseline false positive burden from existing screening. Chat Control would multiply it by orders of magnitude.

The resource reallocation effect operates in two directions simultaneously: it drains investigative capacity toward mass triage of flagged innocent communications, and away from the targeted investigations of known offenders and networks that currently constitute the most effective child protection work. This is not a trade-off between liberty and security; it is a mechanism by which the program would degrade both.

Framing Chat Control as offering such a trade reflects statistical reification (Greenland, 2017): treating hypothetical data distributions and statistical models as if they reflect known physical laws rather than speculative assumptions for thought experiments, and then mapping those distributions onto values without evidence for the leap. Meanwhile, the confusion matrix outputs — true and false positives and negatives — do not map cleanly onto “security gained” and “liberty lost.” Rather, false positives also carry security costs (e.g., finite investigative resources consumed). True positives, likewise, do not guarantee security benefits proportional to their count. Accepting the liberty-versus-security framing tricks policymakers into arguing about whether to accept a trade that is not on the table, while obscuring that the actual choice may instead be between degrading both values or not implementing the program. The scientific evidence does not establish that we are able to trade some liberty for some security here.

The investigative capacity figure (approximately 3,000 EU CSAM investigators, anchored in Europol IOCTA 2023) is the least precisely estimated parameter in this analysis. However, the net harm result is robust to order-of-magnitude variation: even assuming ten times current capacity, false positive triage would consume the full available caseload, leaving Y^* negative across all simulated parameter combinations. A more precise figure has been requested from practitioner sources; current estimates are conservative anchors from published reports.

References cited in this Supporting Information appear in the main-text reference list.